

# SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-17 07:23 UTC

## SEC P vs NP — notebook summary

Generated 2026-05-17 07:23 UTC

- Cycles recorded: **734**
- Time span: 561.7h (~1.31 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp\_notebook.jsonl

### Verdict distribution

| Verdict      | Count |
|--------------|-------|
| INCONCLUSIVE | 696   |
| BARRIER_HIT  | 19    |
| FALSIFIED    | 15    |
| SUPPORTED    | 4     |

### Mathematical fields explored (field\_A)

| Field                                                                             | Cycles |
|-----------------------------------------------------------------------------------|--------|
| Real Algebraic Geometry                                                           | 40     |
| Schur-Weyl Duality                                                                | 28     |
| Free Probability Theory                                                           | 19     |
| Representation Theory of Symmetric Groups                                         | 18     |
| Matroid Theory                                                                    | 16     |
| Free Probability                                                                  | 14     |
| Additive Combinatorics                                                            | 13     |
| Noncommutative $L^p$ Geometry                                                     | 10     |
| Random Matrix Theory                                                              | 8      |
| Polymatroid Theory                                                                | 8      |
| Algebraic Geometry                                                                | 7      |
| Spectral Graph Theory                                                             | 7      |
| Finite Geometry                                                                   | 6      |
| Plethysm Theory                                                                   | 6      |
| Persistent Homology                                                               | 5      |
| Noncommutative Harmonic Analysis                                                  | 5      |
| Ergodic Circuit Framework (communication complexity via dynamical systems)        | 4      |
| {“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”} | 4      |
| Fourier Analysis on Boolean Functions                                             | 4      |

| Field                                                                                              | Cycles |
|----------------------------------------------------------------------------------------------------|--------|
| Algebraic Geometry over Finite Fields                                                              | 4      |
| Fourier Analysis of Boolean Functions                                                              | 4      |
| Algebraic Geometry of Secant Varieties                                                             | 4      |
| FOURIER_ANALYSIS                                                                                   | 4      |
| Noncommutative Geometry                                                                            | 4      |
| Fourier analysis of boolean functions                                                              | 3      |
| Algebraic Topology                                                                                 | 3      |
| matroid theory                                                                                     | 3      |
| Coarse Geometric Lifting (CGL) — Geometric Complexity Theory                                       | 3      |
| {“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”} | 3      |
| Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)                                              | 3      |

### Last 15 cycles (chronological)

| Time                 | Verdict      | Title                                                               |
|----------------------|--------------|---------------------------------------------------------------------|
| 2026-05-17 00:26 UTC | INCONCLUSIVE | 2-adic Newton Slopes of DISJ Sub-blocks Scale as $\sqrt{n}$         |
| 2026-05-17 00:49 UTC | INCONCLUSIVE | Cyclic Burnside Orbit Count Lower-Bounds $AC^0$ Size for PARITY     |
| 2026-05-17 01:28 UTC | INCONCLUSIVE | Cauchy Mean Width of Minterm Hull Lower-Bounds Monotone k-CLIQUE    |
| 2026-05-17 02:01 UTC | INCONCLUSIVE | Costas Displacement Coincidence Lower-Bounds $AC^0$ Size for PARITY |
| 2026-05-17 02:29 UTC | INCONCLUSIVE | Vertex Star-Discrepancy Equals Spectral Norm for XOR Comm           |
| 2026-05-17 02:59 UTC | INCONCLUSIVE | Operator-SoS 4-Trace Gap Lower-Bounds Blocks-Order Read-Twice BP    |
| 2026-05-17 03:06 UTC | BARRIER_HIT  | Cross-Side Fourier Symmetric-Difference Mass Lower-Bounds DISJ CC   |
| 2026-05-17 03:32 UTC | INCONCLUSIVE | q-Major Cancellation Gap Separates Permanent from Determinant Sup   |
| 2026-05-17 04:06 UTC | INCONCLUSIVE | Immanant Positivity Width Bounds Monotone KW Depth for Perfect Ma   |
| 2026-05-17 04:32 UTC | INCONCLUSIVE | Curto-Fialkow Hankel Defect Separates Read-Twice $IP_2$ BPs         |
| 2026-05-17 05:07 UTC | INCONCLUSIVE | Hodge-Cheeger Product Lower-Bounds Tseitin Resolution Length        |

| Time                 | Verdict      | Title                                                                  |
|----------------------|--------------|------------------------------------------------------------------------|
| 2026-05-17 05:45 UTC | INCONCLUSIVE | Sandpile Group Order<br>Lower-Bounds Monotone KW<br>Depth              |
| 2026-05-17 06:17 UTC | INCONCLUSIVE | Conway Temperature Bounds<br>Tree-Resolution Width for<br>3-SAT        |
| 2026-05-17 06:44 UTC | INCONCLUSIVE | Hochster Regularity of<br>Variable Co-Occurrence Graph<br>Lower-Bounds |
| 2026-05-17 07:23 UTC | INCONCLUSIVE | Ihara Zeta Entropy Predicts<br>Tseitin DPLL Tree Size                  |

## How to read the reports

- reports/supported\_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified\_findings.md — FALSIFIED with counterexamples.
- reports/daily\_YYYY-MM-DD.md — chronological log by day.
- pvsnp\_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

## PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
  pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```